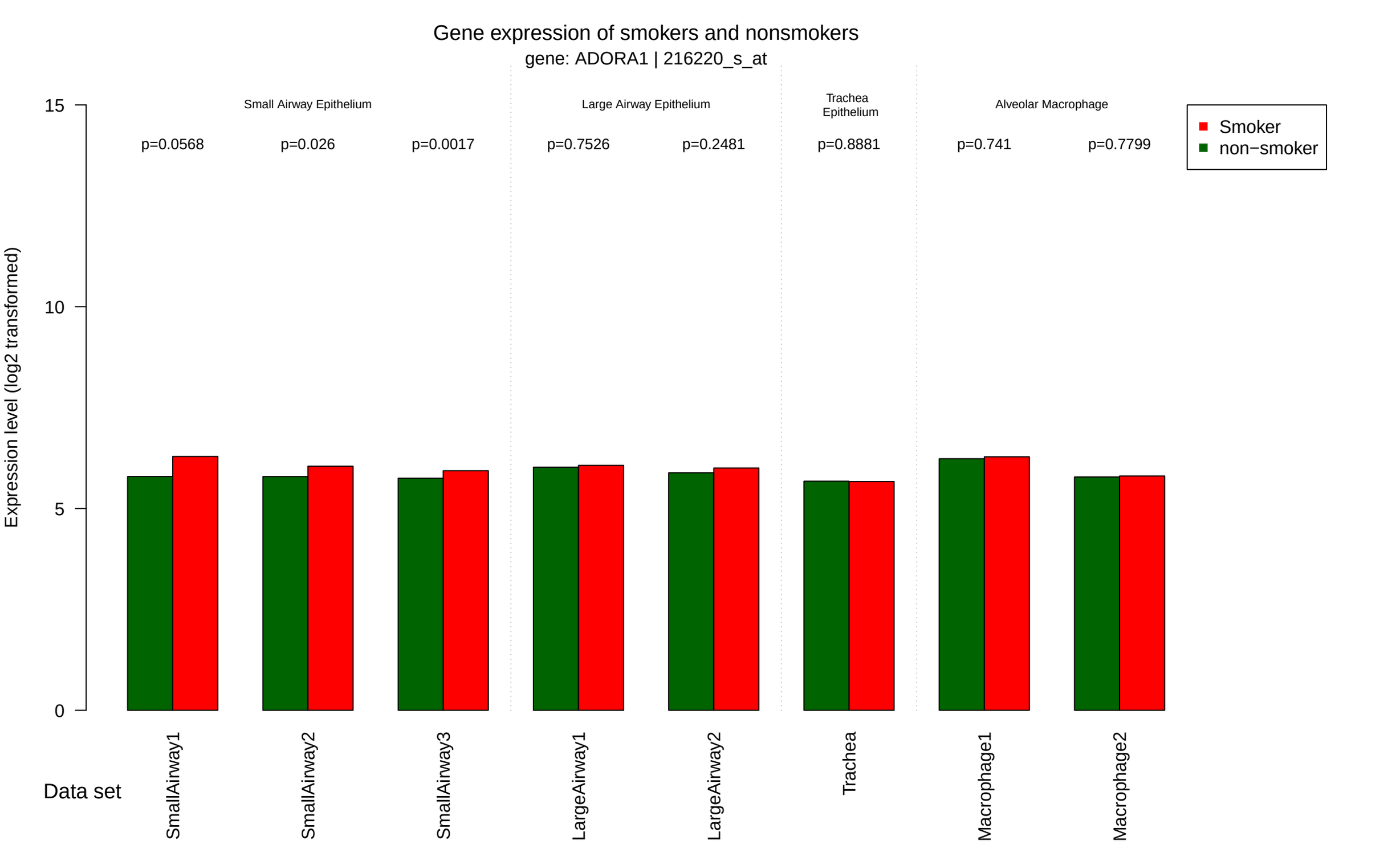
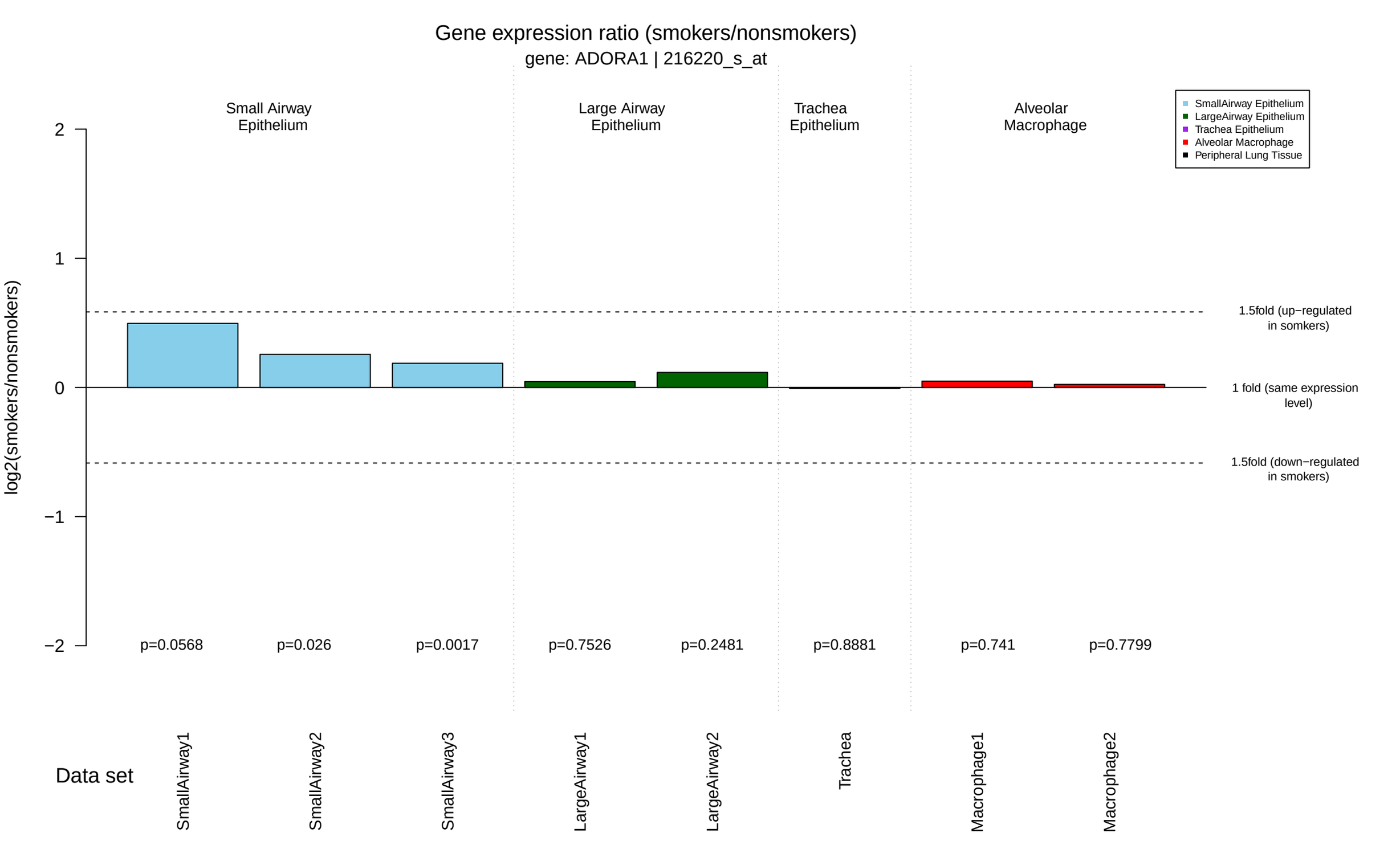
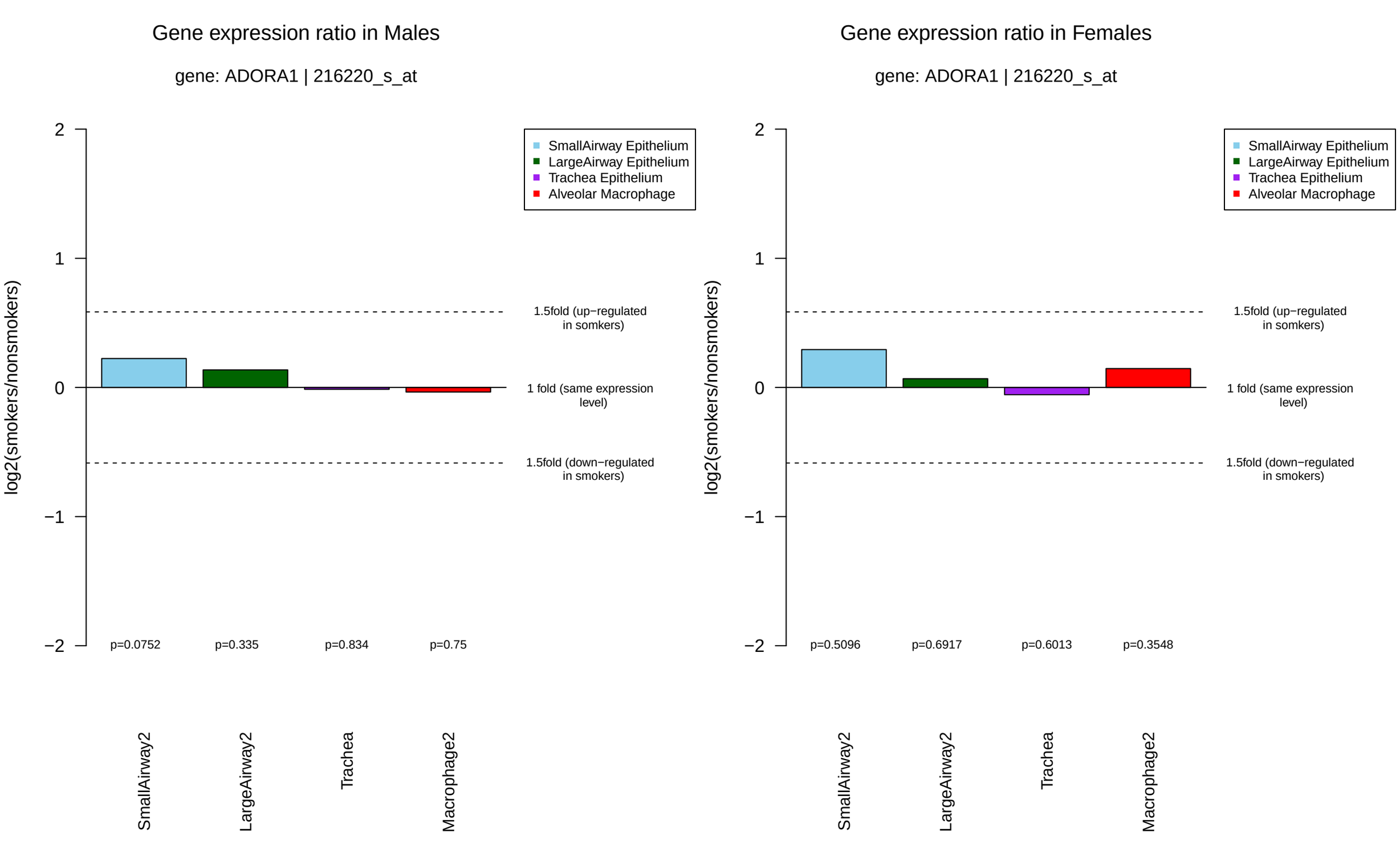
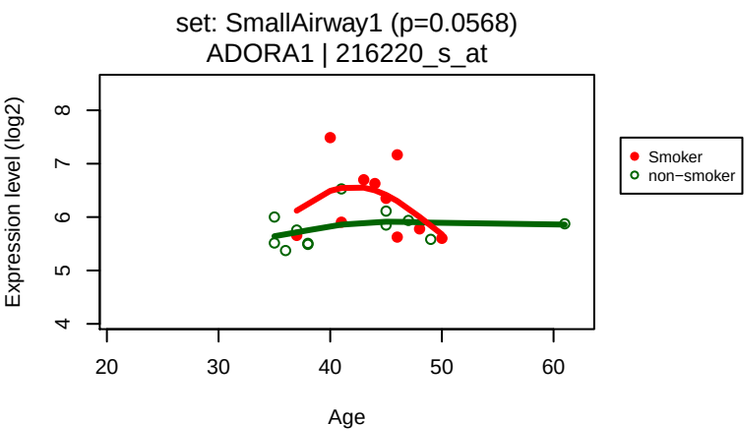
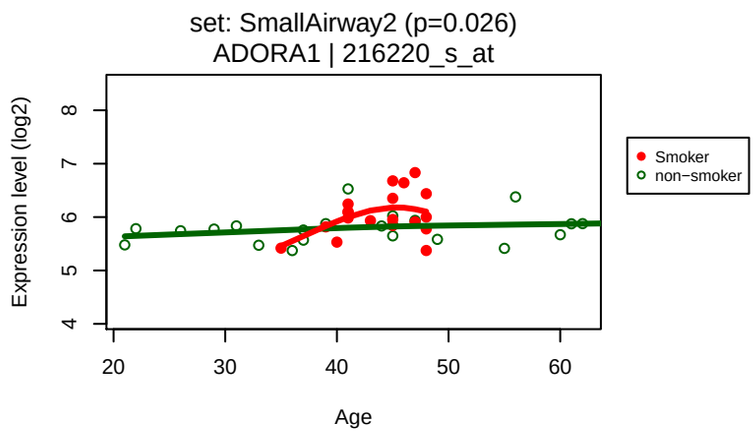
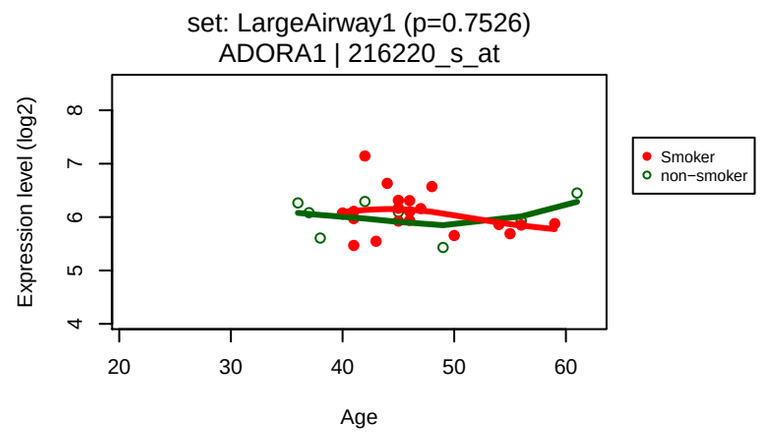

Gene expression of smokers and nonsmokers
gene: ADORA1 | 216220_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0568
p=0.026
p=0.0017
p=0.7526
p=0.2481
p=0.8881
p=0.741
p=0.7799
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

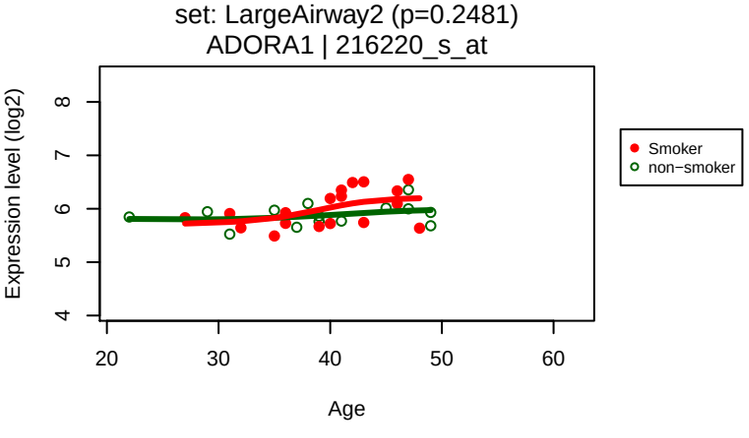

Gene expression ratio (smokers/nonsmokers)
gene: ADORA1 | 216220_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0568
p=0.026
p=0.0017
p=0.7526
p=0.2481
p=0.8881
p=0.741
p=0.7799
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

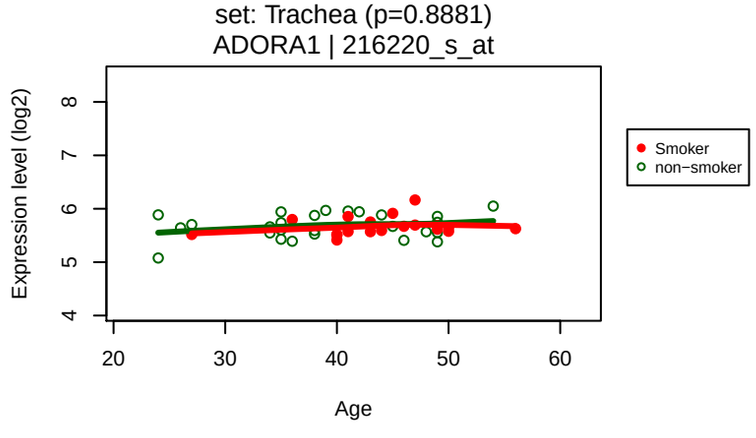

Gene expression ratio in Males
gene: ADORA1 | 216220_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0752
p=0.335
p=0.834
p=0.75
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ADORA1 | 216220_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5096
p=0.6917
p=0.6013
p=0.3548
SmallAirway2
LargeAirway2
Trachea
Macrophage2

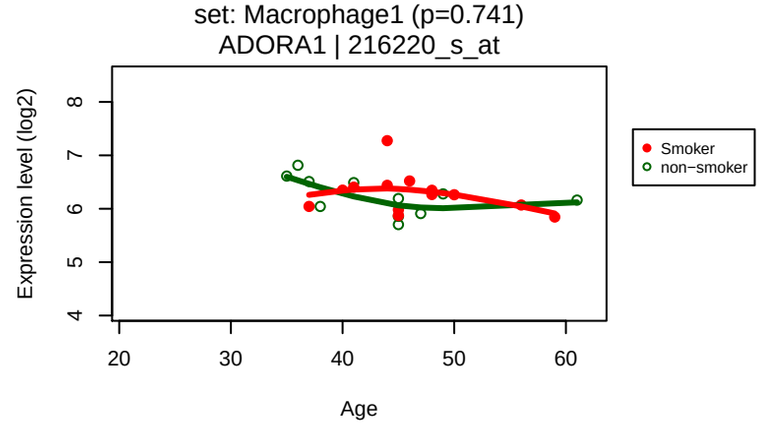

set: SmallAirway1 (p=0.0568)
ADORA1 | 216220_s_at
Expression level (log2)
Age
Smoker
non−smoker

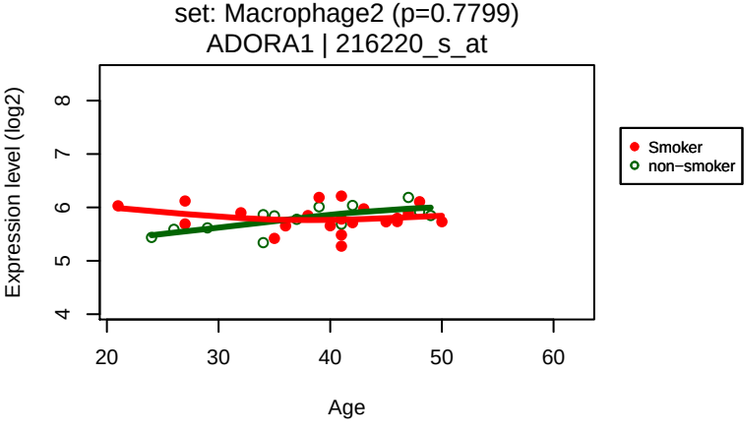

set: SmallAirway2 (p=0.026)
ADORA1 | 216220_s_at
Expression level (log2)
Age
Smoker
non−smoker

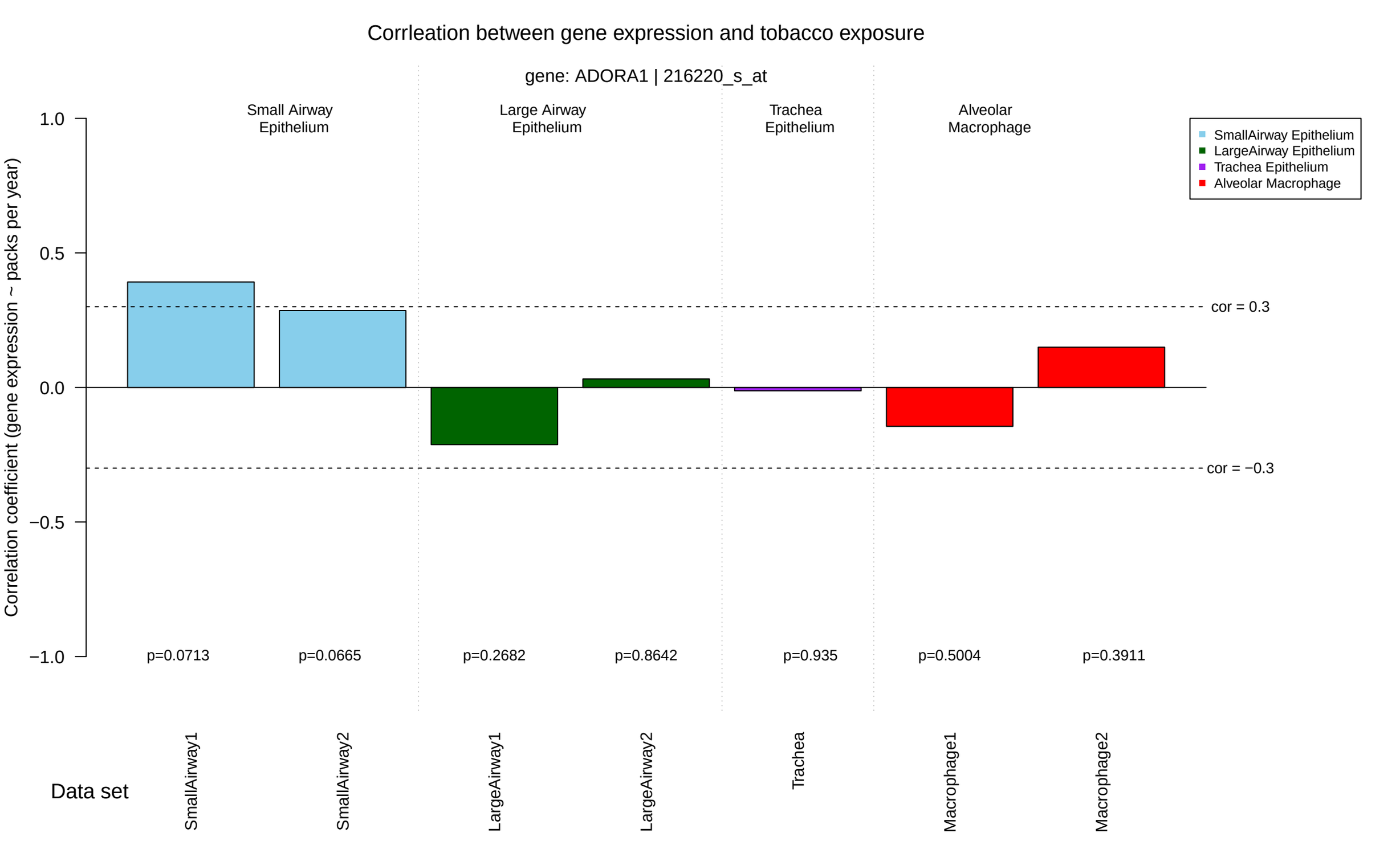

set: LargeAirway1 (p=0.7526)
ADORA1 | 216220_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2481)
ADORA1 | 216220_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8881)
ADORA1 | 216220_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.741)
ADORA1 | 216220_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7799)
ADORA1 | 216220_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ADORA1 | 216220_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0713
p=0.0665
p=0.2682
p=0.8642
p=0.935
p=0.5004
p=0.3911
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2